

HeartGAPSO | Research Data Dossier

Evidence ledger, dataset profile, exact configuration, model comparison and manuscript-ready results | 23 September 2026

Executive finding. The supplied files contain two different hybrid RF experiments. A fast verification run selected 13/13 features; a longer run selected 6/13. The longer run has a separately supplied holdout retest: 51/61 correct (83.61%), ROC-AUC 0.9242, sensitivity 89.29%, specificity 78.79%. Baseline CV scores and the hybrid holdout score use different evaluation protocols, so their ranking is descriptive only.

Evidence classification. A = independently arithmetic-checked against a supplied confusion matrix; B = numerical result transcribed from a run file without raw predictions; C = documentation or cross-protocol comparison requiring rerun. These designations indicate auditability, not clinical validity.

1 | Provenance and reconciliation

Evidence item	Run / interpretation	Status
evaluation(1).json = metrics(1).json	Fast GA-PSO: 13 features; 61 holdout; matrix [[28,5],[2,26]]; AUC 0.9610; RF with 100 trees; threshold 0.50.	A for matrix metrics; B for AUC
optimization_summary(1).json	Fast optimization best fitness 0.820829; confirms 13 features. Runtime 1.91 seconds.	B
optimization_log(1).json + candidate_audit(1).json	Long optimization: 6-feature final candidate; 800 cache misses; 11 logged generations; 3,360 seconds; threshold 0.45 in log.	B
retested_holdout_metrics(1).json + PNGs	Six-feature retest: matrix [[26,7],[3,25]]; ROC-AUC 0.9242; threshold 0.50; matrix and curve PNG match this run.	A for matrix metrics; B for AUC
PIPELINE_RUN.md + Untitled-1.txt	Earlier narrative and full baseline console table. Headline 0.85/0.89 row is a script fallback and conflicts with exported measured JSON.	C for old headline

Key discrepancy: the optimization log identifies the six-feature model at threshold 0.45, while its retest file records threshold 0.50. The files do not establish which saved model was used for retest or why the threshold changed. Preserve both values with their source labels; do not merge them into a single reproducible run.

2 | Dataset, endpoints and preprocessing

Item	Observed value	Interpretation
Source	UCI Heart Disease, Cleveland subset, repository dataset 45	Historical benchmark; no external hospital validation
Samples and predictors	303 records; 13 predictors	Binary target: 0 absent, original 1-4 mapped to present
Class counts	164 absent (54.13%), 139 present (45.87%)	Target balance for complete 303-record dataset
Missingness	ca: 4/303 (1.32%); thal: 2/303 (0.66%)	Median imputation documented; fit in training folds
Holdout	242 development; 61 test; stratified 80/20; seed 42	61-test distribution: 33 absent, 28 present
Preprocessing	Median imputation then MinMax scaling	Requires code-level verification for each model and fold

Predictors: age, sex, chest-pain type (cp), resting BP (trestbps), cholesterol (chol), fasting blood sugar (fbs), resting ECG (restecg), maximum heart rate (thalach), exercise angina (exang), ST depression (oldpeak), slope, major vessels (ca), and thalassemia test category (thal). The source documentation reports no duplicate records. The target is the dataset label, not a prospective clinical outcome.

3 | Methods and model configurations

The pipeline documents a GA candidate with 13 binary feature flags and six Random Forest parameter coordinates. Statistical feature analysis on development data guides initialization/mutation. The fitness objective is 0.40 AUC + 0.25 F1 + 0.20 sensitivity + 0.15 specificity, with separate feature-count and fold-variation penalties as documented. Median imputation and MinMax scaling are specified within training folds. These are implementation claims from supplied documentation; a complete runnable dependency tree was not supplied for independent execution.

Configuration	Fast verification run	Long optimization run
Population / generations	6 / 5	50 / 11 generations logged
PSO iterations / particles	3 / 3	Not fully resolved from log; 220 PSO history entries

Configuration	Fast verification run	Long optimization run
Evaluation budget	30 max fitness evaluations	800 cache misses; 116 cache hits; 512 PSO evaluations
Search timing	1.91 sec	3,360.24 sec (56.00 min)
Selected features	All 13	sex, cp, fbs, slope, ca, thal (6/13)
RF parameters	trees=100; depth=10; split=5; leaf=3; max_features=sqrt; weight=balanced	trees=600; depth=10; split=10; leaf=10; max_features=0.33; weight=None
Training objective	0.820829	0.864525 (final audit); best history 0.871051; different scoring stages
Recorded threshold	0.50	0.45 optimization; 0.50 retest

Search audit issue. The long run mean fitness drops below zero from generation 6 (e.g., -0.8505 at generation 7) while best fitness stays about 0.8711. The log contains candidates with fitness -1. Investigate invalid-candidate handling and the plotted mean before presenting convergence as stable improvement. The fast run fitness curve also descends; it is not evidence of monotonic improvement.

4 | Holdout measurements: recalculated

Metric	Fast RF, 13 features	Long RF, 6 features
Confusion matrix (TN, FP, FN, TP)	28, 5, 2, 26	26, 7, 3, 25
Accuracy	54/61 = 88.52% [78.2-94.3%]	51/61 = 83.61% [72.4-90.8%]
Sensitivity	26/28 = 92.86% [77.4-98.0%]	25/28 = 89.29% [72.8-96.3%]
Specificity	28/33 = 84.85% [69.1-93.3%]	26/33 = 78.79% [62.2-89.3%]
Precision	26/31 = 83.87%	25/32 = 78.13%
F1	52/59 = 88.14%	50/60 = 83.33%
ROC-AUC	0.9610 [exported, no scores provided]	0.9242 [retest file, no scores provided]
False positives / false negatives	5 / 2	7 / 3
Decision threshold	0.50	0.50 in retest file

Square brackets on accuracy/sensitivity/specificity show approximate 95% Wilson binomial intervals calculated from the supplied matrix, not repeated-run intervals. The two rows reuse a reported 61-person test partition, but paired case-level predictions are absent: a paired significance test, AUC uncertainty estimate, or claim that one model is better is not supported.

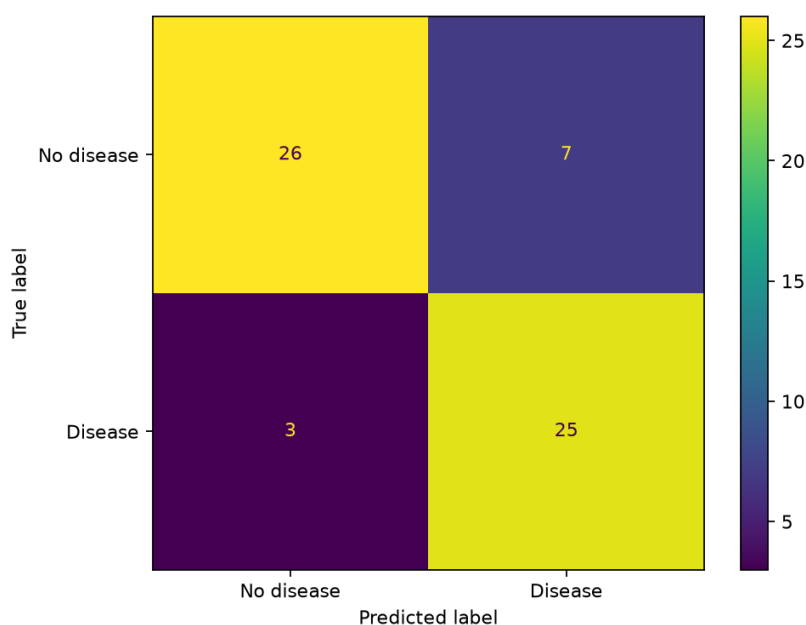


Figure 1. Supplied six-feature retest confusion matrix: true classes in rows, predicted classes in columns.

5 | Baseline comparison: 5x3 nested CV results

The attached console output reports eight classifiers across all features, mutual information k=5, RFECV and RFE k=9, plus two stacking variants. The script passes 5 outer and 3 inner folds. Baselines appear to use the full 303-record dataset, while hybrid numbers above are single holdout results; no direct statistical ranking is valid. Values below are mean +/- reported fold standard deviation, transcribed from the console. RFECV estimator choice varies by model in the shown code despite the generic LR-base label.

Classifier	FS	Accuracy	F1	AUC
LogisticRegression	All	0.8350 +/- 0.0429	0.8137 +/- 0.0502	0.9087 +/- 0.0120
LogisticRegression	MI-5	0.8217 +/- 0.0289	0.8001 +/- 0.0373	0.8927 +/- 0.0215
LogisticRegression	RFECV	0.8284 +/- 0.0338	0.8068 +/- 0.0395	0.9054 +/- 0.0117
LogisticRegression	RFE-9	0.8251 +/- 0.0244	0.8065 +/- 0.0295	0.9009 +/- 0.0195
KNN	All	0.8086 +/- 0.0268	0.7888 +/- 0.0244	0.8746 +/- 0.0140
KNN	MI-5	0.7986 +/- 0.0385	0.7685 +/- 0.0491	0.8494 +/- 0.0336
KNN	RFECV	0.8152 +/- 0.0315	0.7937 +/- 0.0374	0.8594 +/- 0.0079
KNN	RFE-9	0.8121 +/- 0.0391	0.7897 +/- 0.0452	0.8740 +/- 0.0250
NaiveBayes	All	0.8546 +/- 0.0403	0.8385 +/- 0.0416	0.9022 +/- 0.0230
NaiveBayes	MI-5	0.8118 +/- 0.0430	0.7901 +/- 0.0452	0.8662 +/- 0.0295
NaiveBayes	RFECV	0.8250 +/- 0.0360	0.8067 +/- 0.0342	0.8974 +/- 0.0185
NaiveBayes	RFE-9	0.8448 +/- 0.0429	0.8254 +/- 0.0500	0.8959 +/- 0.0132
DecisionTree	All	0.7786 +/- 0.0678	0.7552 +/- 0.0784	0.8091 +/- 0.0690
DecisionTree	MI-5	0.7950 +/- 0.0751	0.7623 +/- 0.0821	0.8237 +/- 0.0648
DecisionTree	RFECV	0.7590 +/- 0.0657	0.7374 +/- 0.0682	0.7983 +/- 0.0710
DecisionTree	RFE-9	0.7852 +/- 0.0713	0.7611 +/- 0.0815	0.8128 +/- 0.0675
SVM	All	0.8283 +/- 0.0478	0.8079 +/- 0.0535	0.8973 +/- 0.0257
SVM	MI-5	0.8152 +/- 0.0191	0.7926 +/- 0.0306	0.8773 +/- 0.0246
SVM	RFECV	0.8152 +/- 0.0383	0.7968 +/- 0.0426	0.8912 +/- 0.0161
SVM	RFE-9	0.7987 +/- 0.0435	0.7781 +/- 0.0488	0.8832 +/- 0.0216
RandomForest	All	0.8249 +/- 0.0492	0.8001 +/- 0.0541	0.9057 +/- 0.0231
RandomForest	MI-5	0.8184 +/- 0.0161	0.7923 +/- 0.0284	0.8844 +/- 0.0317
RandomForest	RFECV	0.8349 +/- 0.0363	0.8087 +/- 0.0423	0.9098 +/- 0.0254
RandomForest	RFE-9	0.8282 +/- 0.0380	0.8034 +/- 0.0479	0.8954 +/- 0.0262
ExtraTrees	All	0.8381 +/- 0.0392	0.8222 +/- 0.0449	0.9103 +/- 0.0225
ExtraTrees	MI-5	0.8217 +/- 0.0289	0.8001 +/- 0.0373	0.8841 +/- 0.0217
ExtraTrees	RFECV	0.8348 +/- 0.0399	0.8152 +/- 0.0463	0.9125 +/- 0.0192
ExtraTrees	RFE-9	0.8383 +/- 0.0194	0.8158 +/- 0.0258	0.9041 +/- 0.0197
XGBoost	All	0.8019 +/- 0.0385	0.7816 +/- 0.0467	0.8928 +/- 0.0193
XGBoost	MI-5	0.8182 +/- 0.0443	0.7920 +/- 0.0508	0.8793 +/- 0.0396
XGBoost	RFECV	0.8053 +/- 0.0119	0.7834 +/- 0.0157	0.8805 +/- 0.0239
XGBoost	RFE-9	0.8115 +/- 0.0465	0.7861 +/- 0.0586	0.8946 +/- 0.0135
StackedEnsemble_NoFS	All	0.8250 +/- 0.0272	0.8093 +/- 0.0234	0.9107 +/- 0.0197
StackedEnsemble_WithFS	RFE-9	0.8250 +/- 0.0314	0.8080 +/- 0.0273	0.8923 +/- 0.0299

34 unique configurations extracted. Two stacking configurations also appear twice in the console summary and were deduplicated here. A separate exploratory MI k sweep in PIPELINE_RUN.md reports k=5,7,9,11,13; it is a tuning observation, not 40 additional uniformly documented model results. Best documented MI sweep: RF at k=9, AUC 0.9128; original console grid above lists MI k=5 only.

6 | Model configuration grids and comparison rules

Model	Documented candidate grid / configuration
Logistic regression	C: 0.001, 0.01, 0.1, 1, 10, 100; penalty: l1, l2
KNN	Neighbors: 3, 5, 7, 9; weights: uniform, distance
Gaussian naive Bayes	No grid documented
Decision tree	Depth: 3, 5, 7, None; min split: 2, 5, 10

Model	Documented candidate grid / configuration
SVM	C: 0.1, 1; kernel: linear, RBF (parameter prefix estimator__ in notebook)
Random forest	Trees: 50, 100, 200; depth: 5, 10, None
Extra Trees	Trees: 50, 100, 200; depth: 5, 10, None
XGBoost	Trees: 50, 100, 200; depth: 3, 5, 7; learning rate: 0.01, 0.1, 0.2
Stacked ensemble	Base models from registry; LogisticRegression meta-learner, liblinear; no separate grid in shown call
GA-PSO tuned RF	Fast and long winning configurations reported in section 3; full search-space options in CONFIGURATION.md

These are declared search grids in the supplied monolithic script, not independently verified winning configurations per baseline. The registry implementation was not supplied in this attachment set. The manuscript should report selected per-fold parameters only after exporting them from the actual CV results.

7 | Optimization audit and plots

The long optimizer log contains 11 generation records, 800 evaluations, 116 cache hits, 512 PSO evaluations, 55 high-fidelity evaluations, 30 audited candidates and 220 PSO history records. The highest ranked audited candidate uses six features and 600 RF trees. It has final fitness 0.864525; its internal CV AUC from fitness_metrics is 0.9039. These are training-side metrics.

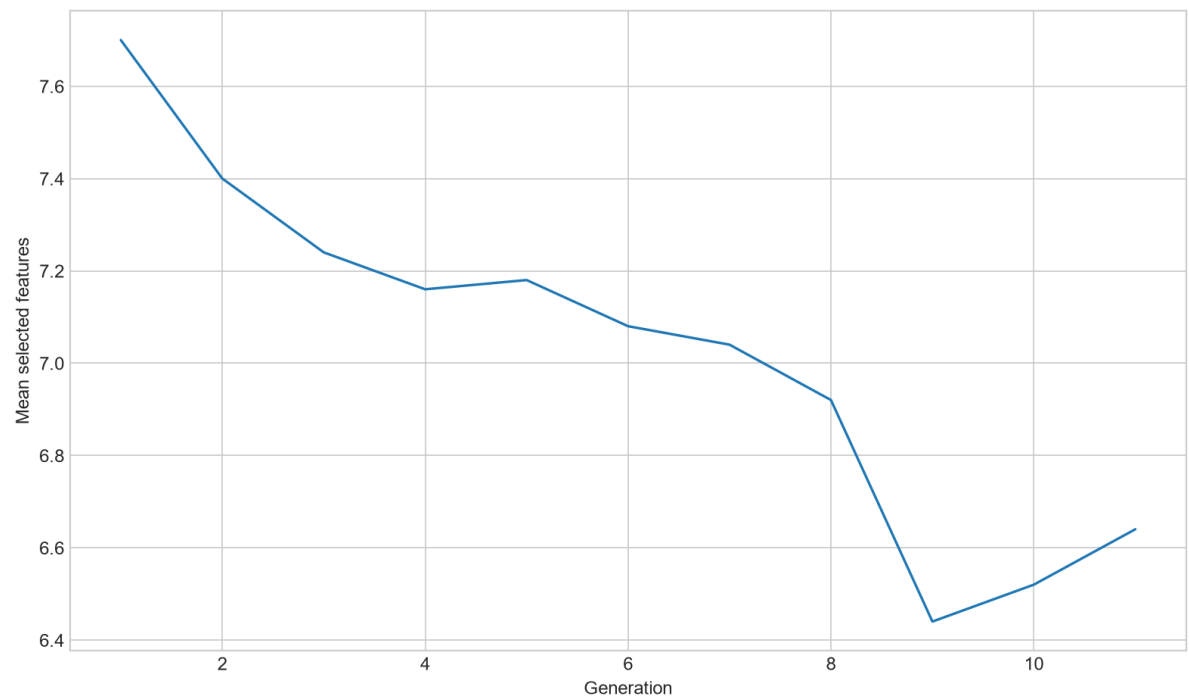


Figure 2. Mean selected features per generation (long optimization); this does not show test-set feature importance.

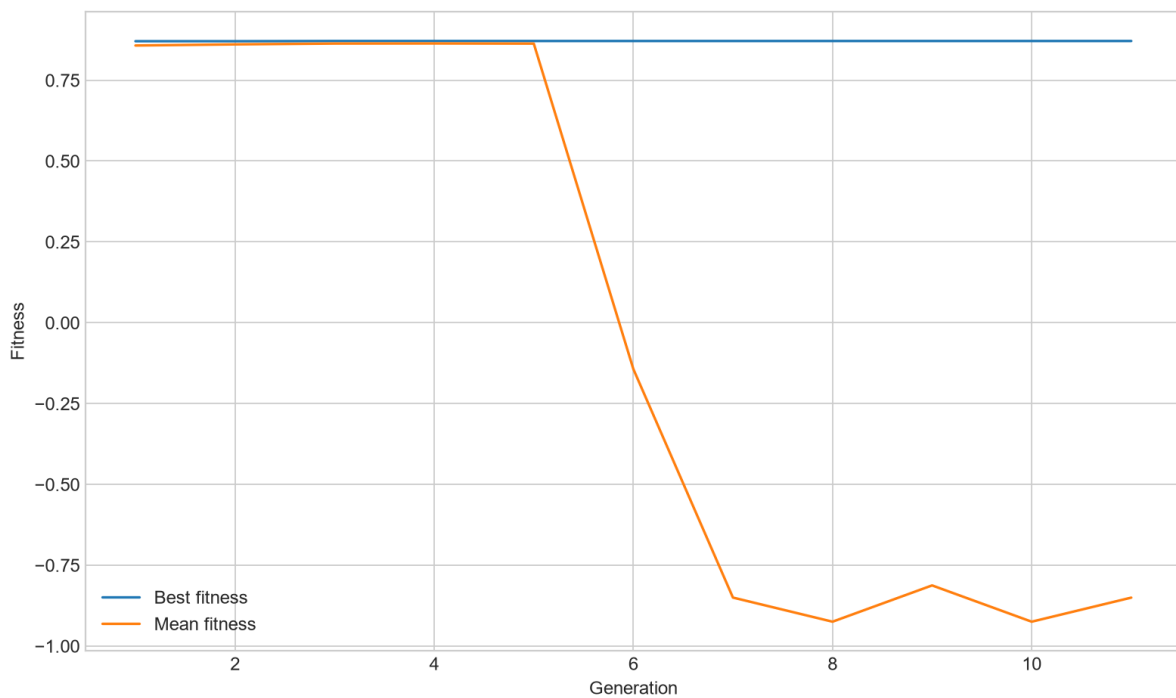


Figure 3. Best and mean fitness, long optimization. Negative mean values warrant investigation.

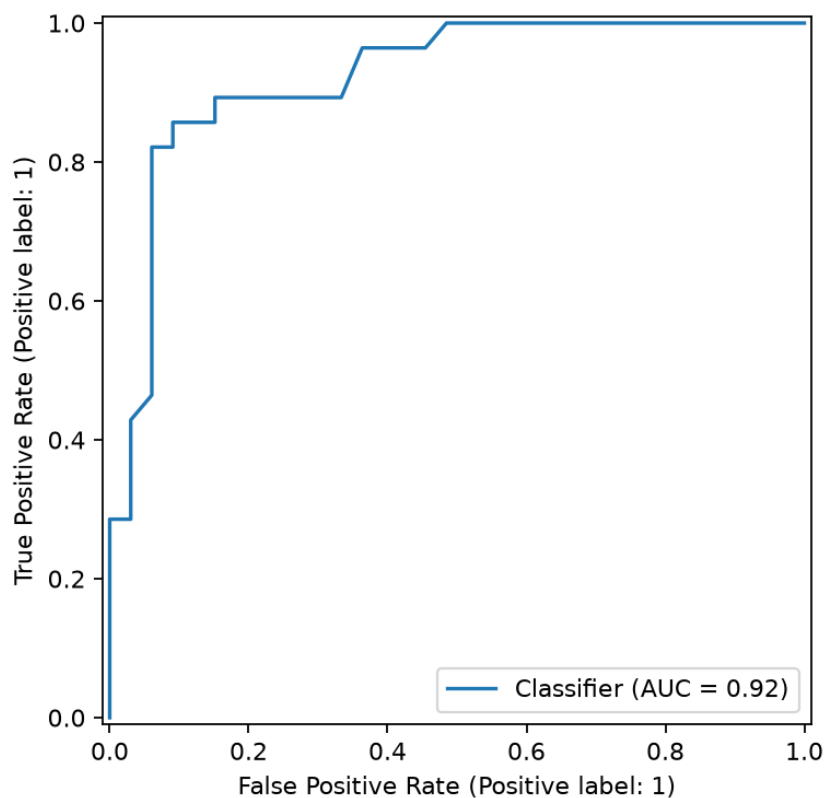


Figure 4. Supplied six-feature retest ROC curve; AUC shown rounded to 0.92.

8 | Related work and interpretation

El-Shafiey et al. (2022), "A hybrid GA and PSO optimized approach for heart-disease prediction based on random forest," Multimedia Tools and Applications 81:18155-18179, doi:10.1007/s11042-022-12425-x, report Cleveland accuracy of 87.8% using 10-fold CV and 95.6% using five 70/30 holdout repetitions. Their splits and repeated evaluation differ from both local experiments. Treat those values as historical context, not direct superiority evidence. The GA-PSO-RF combination and Cleveland application are prior work; articulate the new optimizer audit, joint tuning, threshold handling or rigor as the actual study contribution, conditional on reproducible experiments.

9 | Paper-ready result text (qualified)

"On the 303-record Cleveland dataset, the six-feature GA-PSO selected random forest was evaluated on a reported 61-record stratified holdout. The retest file records 51 correct classifications (83.61%; approximate 95% Wilson interval 72.4%-90.8%), sensitivity of 25/28 (89.29%), specificity of 26/33 (78.79%), precision of 25/32 (78.13%), F1 of 0.8333, and a reported ROC-AUC of 0.9242. The selected variables were sex, chest-pain type, fasting blood sugar, ST-slope, number of major vessels, and thal category. Because per-person prediction scores and the exact saved model provenance were unavailable, the reported AUC and model identity require independent reproduction."

10 | Minimum reproducibility package before submission

Priority	Required export or correction	Why
1	Patient-level test rows: index, true label, probability, predicted label for both models	Recalculate AUC; paired comparison; calibration; error analysis
2	Hash and save exact train/test indices and preprocessing pipeline; record dataset checksum	Confirm same untouched holdout and no leakage
3	Link retest model artifact and configuration to optimizer log; explain 0.45 vs 0.50 threshold	Resolve six-feature run identity
4	Repeat all models on identical outer folds with feature selection inside each fold; save fold metrics	Fair comparison and confidence intervals
5	Add GA-only, PSO-only, untuned RF and simple LR ablations on same protocol	Demonstrate added value of hybrid components
6	Inspect generation 6-11 negative fitness and audit score-stage differences	Avoid misleading convergence assertions
7	Report discrimination, calibration and intended-use limitations under TRIPOD+AI	Clinical prediction reporting completeness

Sources and data boundary

Supplied artifacts: PIPELINE_RUN.md; METHODOLOGY.md; ALGORITHM.md; CONFIGURATION.md; API_REFERENCE.md; Untitled-1.txt; heartgapso_pipeline_v4 (3)(2).py; evaluation(1).json; metrics(1).json; optimization_summary(1).json; optimization_log(1).json; candidate_audit(1).json; retested_holdout_metrics(1).json; supplied PNG figures; reference article PDF. Original patient-level CSV, exact model binary, test probabilities and complete independent baseline registry were not attached.

UCI Machine Learning Repository, Heart Disease dataset: <https://archive.ics.uci.edu/dataset/45/heart+disease> . TRIPOD+AI statement: Collins et al., BMJ 2024;385:e078378; <https://doi.org/10.1136/bmj-2023-078378> . El-Shafiey et al. 2022: <https://doi.org/10.1007/s11042-022-12425-x> .